

The empirical recurrence for OEIS A305520, recovered from its tail

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Abstract

OEIS A305520 records “Empirical recurrence of order 58” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 117$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 60$. Its last coefficient is $c_{58} = 21440 \neq 0$, so no further tail satisfies anything shorter; any order-58 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A305520 is “Number of $n \times 5$ 0..1 arrays with every element unequal to 0, 1, 2, 3, 4 or 5 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

16, 512, 10082, 186237, 3536750, 66949063, 1269418258, 24080732454, ...

The entry states:

Empirical recurrence of order 58 (see link above)

As of the “Last modified” line on the live entry (Jun 04 2018 11:54:48, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 117$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 117$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 58: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 58.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 234$ exact terms from $n = 2$ on, it returns order 58 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{58} c_i a(n-i),$$

c_1	16	c_{16}	−3730380978	c_{31}	−41647751483	c_{46}	−10014686715
c_2	140	c_{17}	−1893030283	c_{32}	−128450542684	c_{47}	−6470578927
c_3	−1032	c_{18}	5741197383	c_{33}	−57720901695	c_{48}	−2720603649
c_4	−11242	c_{19}	12446048567	c_{34}	88925281354	c_{49}	−515496268
c_5	−2564	c_{20}	10300431036	c_{35}	185093796404	c_{50}	−26222713
c_6	273498	c_{21}	−2070162693	c_{36}	173035674811	c_{51}	4984968
c_7	973185	c_{22}	−8505117328	c_{37}	79741601317	c_{52}	−47276447
c_8	−1228562	c_{23}	4629174421	c_{38}	−6628188618	c_{53}	−55659916
c_9	−14145257	c_{24}	16853452669	c_{39}	−54454323296	c_{54}	−4240340
c_{10}	−24404896	c_{25}	−4990948544	c_{40}	−45416773591	c_{55}	2365374
c_{11}	39686100	c_{26}	−70271940387	c_{41}	−8888880012	c_{56}	−4244852
c_{12}	239434776	c_{27}	−90719572547	c_{42}	6272135071	c_{57}	−1096354
c_{13}	410279999	c_{28}	−11111464918	c_{43}	5894030130	c_{58}	21440
c_{14}	−198135616	c_{29}	90468144376	c_{44}	−3050282937		
c_{15}	−2059823507	c_{30}	86180403566	c_{45}	−10613477334		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 60$, and it is the recurrence OEIS A305520 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{58} - c_1 x^{57} - \dots - c_{58}$. Its constant term is $-c_{58} = -21440$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 58 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 58, so $\deg q = 58$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 58, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 15 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 58, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 21440.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-58 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A305520.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.